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B.C-CSE

CO2-AT2 ACSA1603- DWDM

1. (a) Define data mining & explain the KDD process.

Data mining:

Data Mining is the process of discovering useful pattern, relationships, and trends from large amounts of data using statistical and M-L techniques

Two basic data mining tasks:

1. classification:
   Assigns data to predefined categories.

2. clustering:
   Groups similar data items based on their characteristics.

Steps involved in KDD process:

| |
|---|
| Data selection |
| Data preprocessing |
| Data transformation |
| Data mining |
| Pattern evaluation |

(b) What is data cleaning? Explain data quality issues and methods to handle missing and noisy data.

Data cleaning:
Is the process of identifying and

correcting or removing inaccurate, incomplete, duplicate and inconsistent data.

two common data quality issues:

1. Missing values
2. Noisy data.

Methods to handle missing values:

↳ Replace with mean, median or mode

↳ Remove records when appropriate.

↳ estimate values using prediction methods to handle noisy data:

Binning:

smooth values using binnary or medians

Regression:

Fit a function of the data.

outlier detection:

Identify unusual values & verify them.

2(a). Define data integrations & explain its challenges

Data integration combines data from multiple sources into one consistent database

Two challenges:

schema differences: Different column for the same attribute.

Data conflicts:

different values for the same record.

Handling redundancy: Identify duplicate records and remove unnecessary copies.

(b) What is covariance analysis? Just 2 formulaes.

Covariance analysis measures how 2 variable change together.

→ positive covariance: variables tend to move in the same direction.

→ negative covariance: variables tend to move in opposite direction.

→ zero variance: No linear covariance as

1. popular covariance:

$$cov(x,y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$$

2. sample covariance:

$$S_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

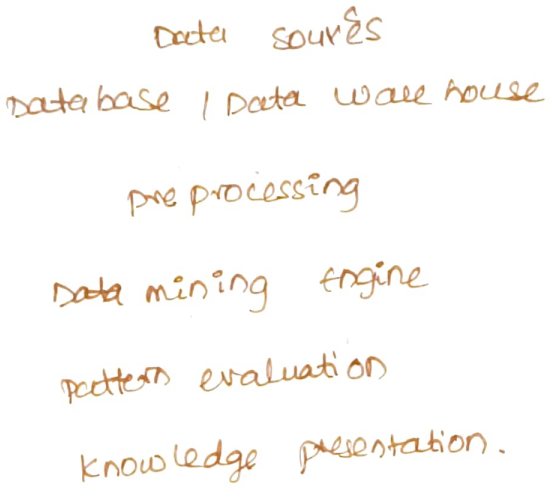
3. Analyze the relationship between stocks A & B

Given (2,5)(3,8),(5,10)(4,11)(6,14)

Aim:

* The population covariance is 4.2

Diagram:




components:

Data sources: provide raw data

Database/warehouse: stores and manages data.

pre processing: clean and integrates data

data mining engine: Discovers useful patterns

pattern evaluation: Identifies investing patterns.

4. (a) find covariance for a given data set

Given $x = \{4,5,6,7,9\}$

$y = \{8,3,7,5,6\}$

$\bar{x} = 31/5 = 6.2$

$\bar{y} = 29/5 = 5.8$

$= \Sigma (x_i - \bar{x})(y_i - \bar{y}) \div n$

$= cov(x,y) = \frac{-1.8}{5}$

$cov(x,y) = -0.36$

(b) Major steps:

Data cleaning:

handles missing values errors, noise

Data integration:

continues data from multiple sources

Data integration sources: convert data into suitable format

Data transformation:

convert data into suitable format

5. A perform equal-frequency mining on the age dataset

Given:

27 age values

No. of bins = 3

value per bin = $\frac{27}{3} = 9$

↳ Bin 1

13, 15, 16, 16, 19, 20, 20, 21, 22

Mean = 18

↳ Bin 2

22, 25, 25, 25, 30, 33, 33, 35

Mean = 28.1

↳ Bin 3:

35, 35, 35, 36, 40, 45, 46, 52, 70

Mean = 43.78

Data discretization:

The process of converting continous numerical data into a limited number of intervals or categories

Importance:

* Simplifies data
* Reduces complexity
* Helps data mining algorithms identify
* patterns.

| Equal-width Binning | Equal-frequency Binning |
| --- | --- |
| Divides the data range into equal-width intervals | Divides data bins width appear equal no. of values |
| Interval width = [max - min] / No. of bins | each bin contains approx. the same no. of records |
| Sensitive to extreme values. | Bin width may differ |
| eg: ages<br>10-20, 20-30, 30-40 | eg: 27 values divided into 3 bins of a values. |